

Lab 4: Photodiodes

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This lab investigates the behavior of photodiodes. A photodiode is a semiconductor that converts light into electrical energy using the photoelectric effect. The standard photodiode consists of a p-n junction. Meaning it contains a p-type region and an n-type region. When a photon is captured and contains more energy than the bandgap, it gets absorbed into the depletion region creating an electron-hole pair. The device can be operated in forward and reverse-bias too. In forward bias, it acts like a solar cell, generating a voltage if a load is connected. In reverse-bias, its depletion region widens and has a lower capacitance making it easier to generate current, but it also drastically increases noise.

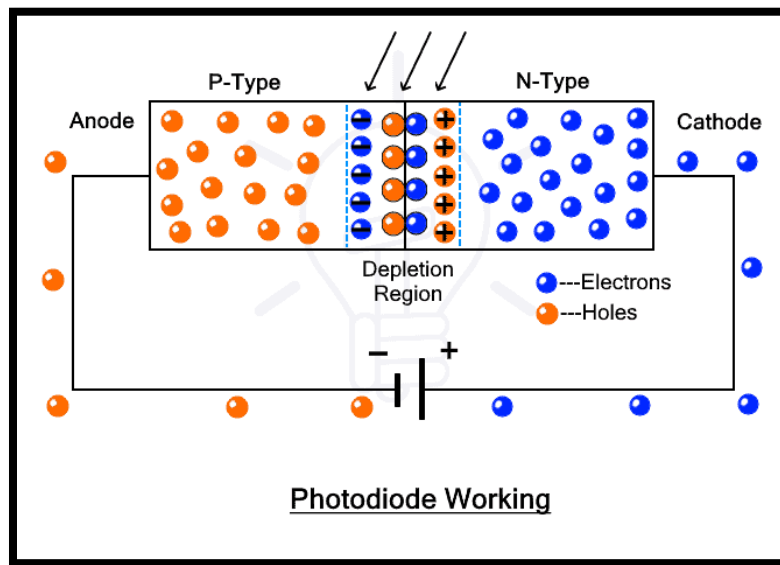


Figure 1: Structure of a Photodiode

Fundamentals:

A silicon-based PIN photodiode has a much larger and more doped N-type region, a less doped intrinsic region, and a smaller but heavier doped p-type region compared to a normal photodiode. Similar to a normal photodiode, it still contains a junction with a depletion region. When high energy light is absorbed and fits into the necessary wavelength requirement, an electron-hole pair is created. The same as explained in the device background. The depletion region for a PIN photodiode also has its own electric field. This field affects the carriers and generates a current in relation to the incident power. The forward-bias and reverse-bias behavior remains the same for a PIN diode.

The most important aspect of a photodiode is its efficiency, η . The higher efficiency, the more current created from an incident light source. The dark current is the reverse-bias current when there is no light. The higher the dark current means there is more noise. Also meaning that weaker signals will get lost in the noise and that the diode will saturate earlier.

The nanohub pnjunctionlab simulation software is set up for a PIN photodiode. Simulations are run for the device under 0V and -5V applied bias. Plots are created for the energy band diagram and the electrostatic potential of the device. The parameters used:

- P-type Length: 0.3 μm N-type Length: 0.3 μm
- P-type Nodes: 30 N-type Nodes: 30
- Intrinsic Length: 10 μm
- Intrinsic Nodes: 400

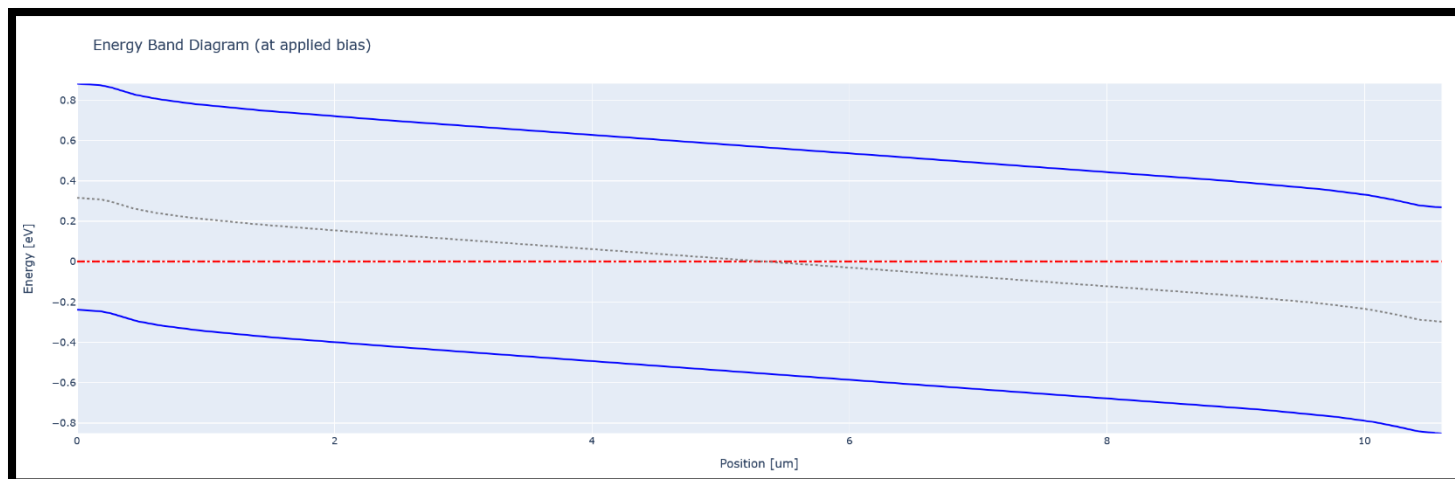


Figure 2: 0V Applied Energy Band Plot

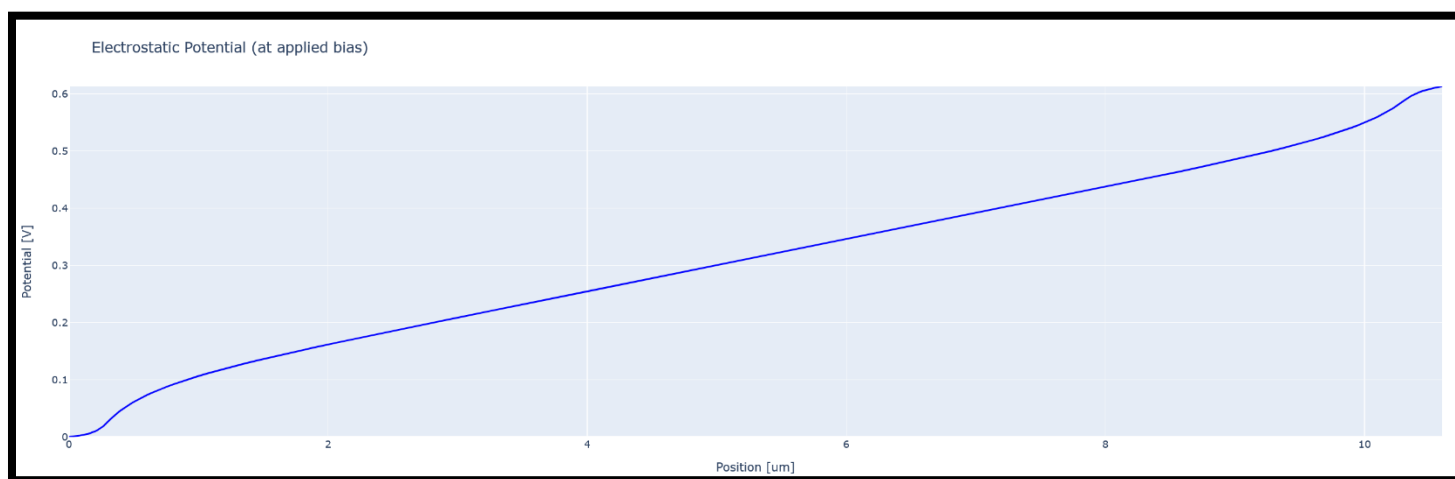


Figure 3: 0V Electric Potential Plot

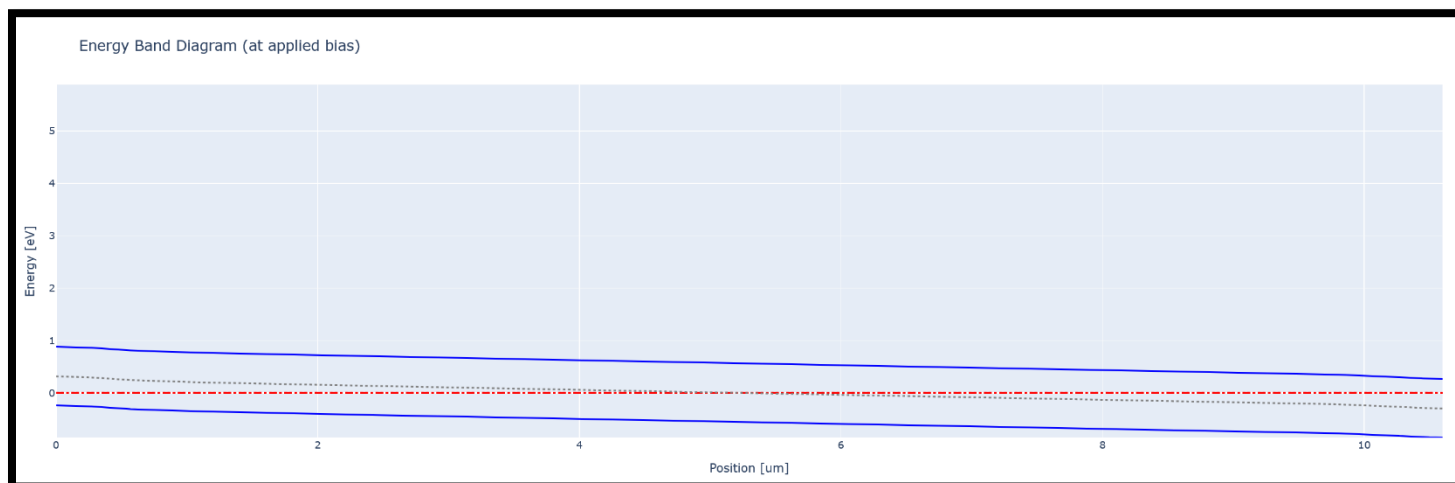


Figure 4: -5V Applied Energy Band Plot

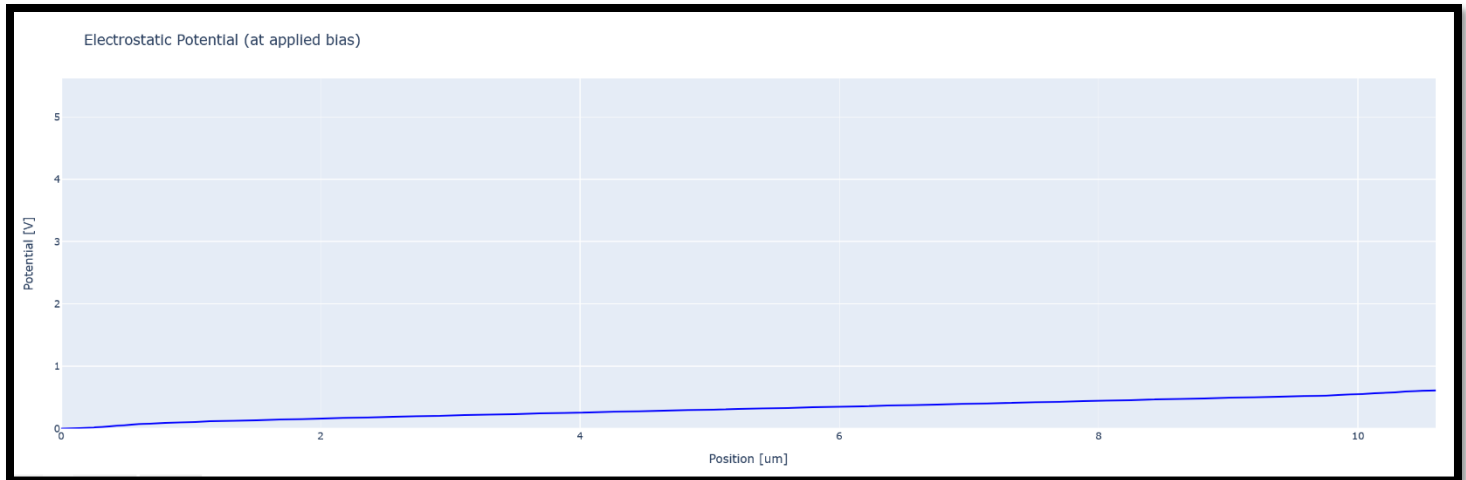


Figure 5: -5V Electric Potential Plot

Experiment 1: I-V Properties and Negative-Bias Photoresponse of the Photodiode:

The goal of experiment 1 is to measure the I-V curve of the photodiode in the dark and from -5V up to 1.5V (just past the turn on voltage) with a blue LED at a constant current. The photodiode used for this experiment is the BPW34. A Keysight U2722A SMU and its AMM software is utilized to run a sweep of the BPW34 to record measurements. This significantly speeds up the time. The setup is simple. The photodiode is connected to a breadboard and wired to the SMU. A cardboard box is placed over the circuit to block any light. The SMU is used to measure the sweep from 0V to 1.5V. To ensure enough datapoints, a setup size of 0.05V is set. The datapoints are analyzed and fit to the non-ideal diode equation. After measuring in the dark, the setup is repeated but this time a blue LED is connected to the breadboard and angled over the BPW34. The LED is connected to the second terminal of the SMU to run at 5 various constant currents (the changing light intensity). The following five currents are what were tested: 20.05 mA, 15.90 mA, 6.68 mA, 1.13 mA, and 0.02 mA. For method E and G, the negative voltage data from -5V to 0V for each light intensity did not get recorded. It is believed this occurred due to a

setup error using the SMU's software. A different lab groups data was patched in for the -5V to 0V parts of the curves. As well as used for part G, as we cannot analyze our own.

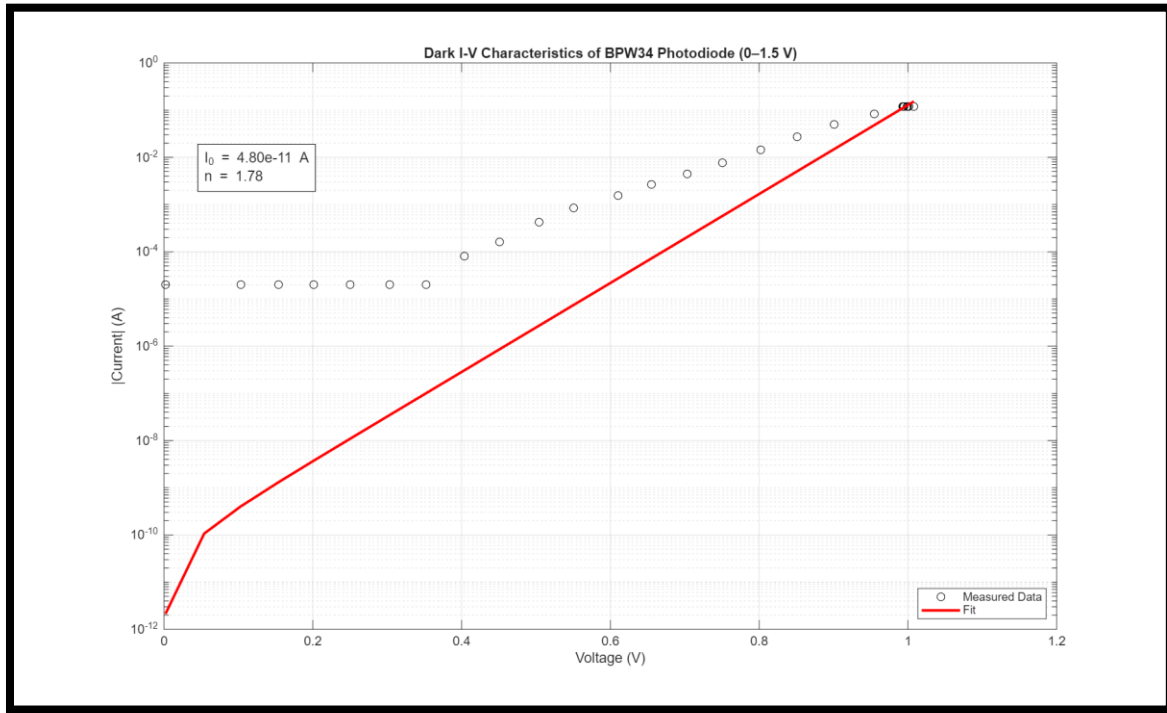


Figure 6: Experiment 1 Method A Dark Current Fit to Non-Ideal Diode Equation

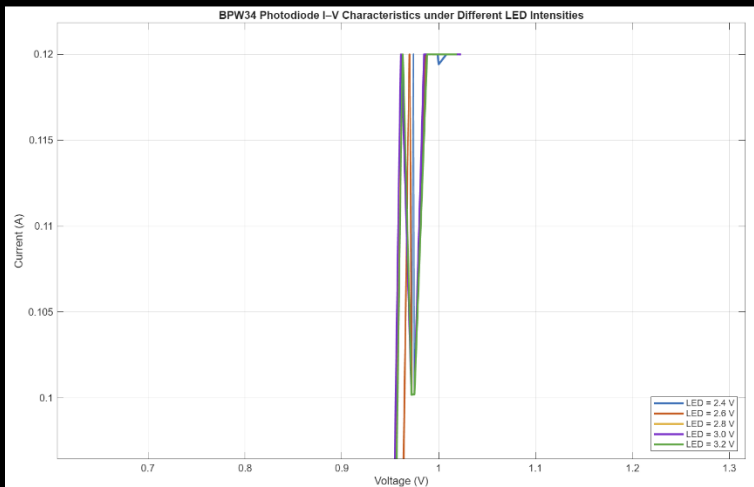
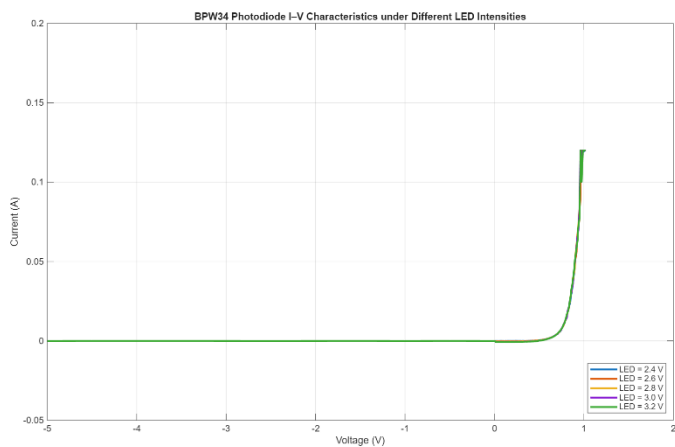


Figure 7 and 8: Experiment 1 Method E Different Light Intensities

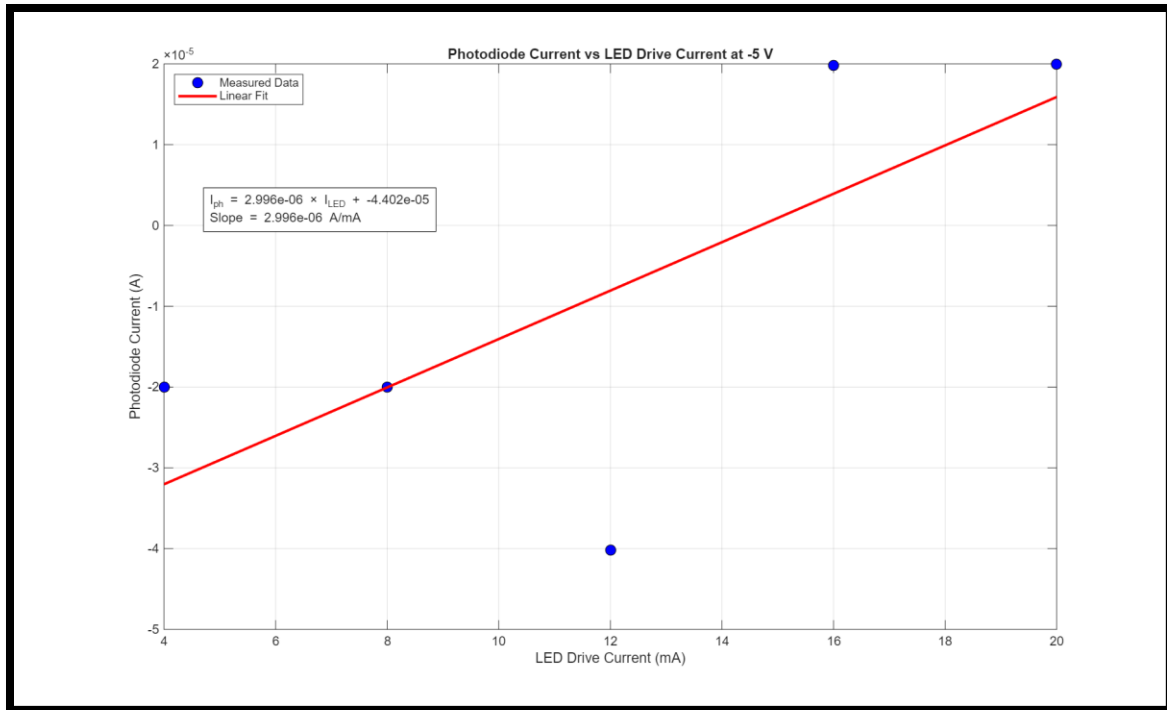


Figure 9: Experiment 1 Method G Photodiode Current at -5V for Each Light Intensity

The non-ideal diode fit could have been better but does not stray far off from the linear/exponential part of the curve. The fit yielded an n value of 1.78 and an I_0 of 4.8×10^{-11} A as shown in Figure 6. When powering the LED with different light intensities, the data yielded extremely similar values. Even with the plot zoomed in, the final voltages of each different light intensity value ended extremely close to the previous. When extracting the photodiode current at -5V and comparing it to the different light intensities and drive current, it is clear there is a linear relationship.

Concept Questions:

According to my recorded fit, this particular photodiode has a reasonable n value of 1.78. Compared to previous labs, it is far more ideal than others. It decently maintains linearity when in the dark and when compared with different light intensities. There are a few discrepancies seen at higher intensity levels, but this is most likely due to dark-current and noise.

Experiment 2: Investigating the Smallest Light Intensity the Photodiode can Detect

The goal of experiment 2 is to measure the photodiode's response to a signal noise of 1:1. This will help determine a measurement for the smallest intensity that the photodiode can detect. Similar to experiment 1, a blue LED is angled over the BPW34 photodiode. The photodiode is wired to an oscilloscope to measure the voltage. A waveform generator is connected to the blue LED with a 7V peak-to-peak amplitude, 1 kHz frequency, and 50% duty cycle for power. On the oscilloscope's second channel, this wave is measured. With the scopes captured, the peak-to-peak voltage of the LED's wave is lowered until the signal-to-noise ratio is $\sim 1:1$.

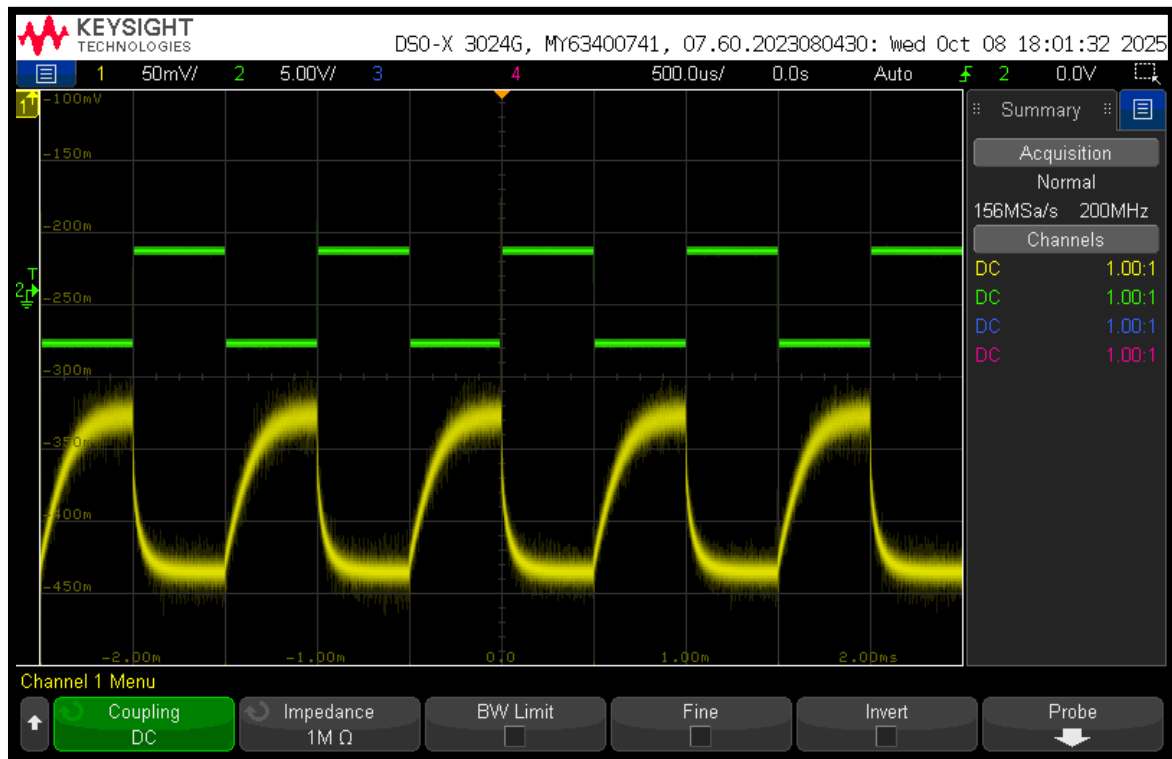


Figure 10: Green: Input Square Wave 1 kHz, 7V pk-to-pk, 50% duty cycle Yellow: Photodiode

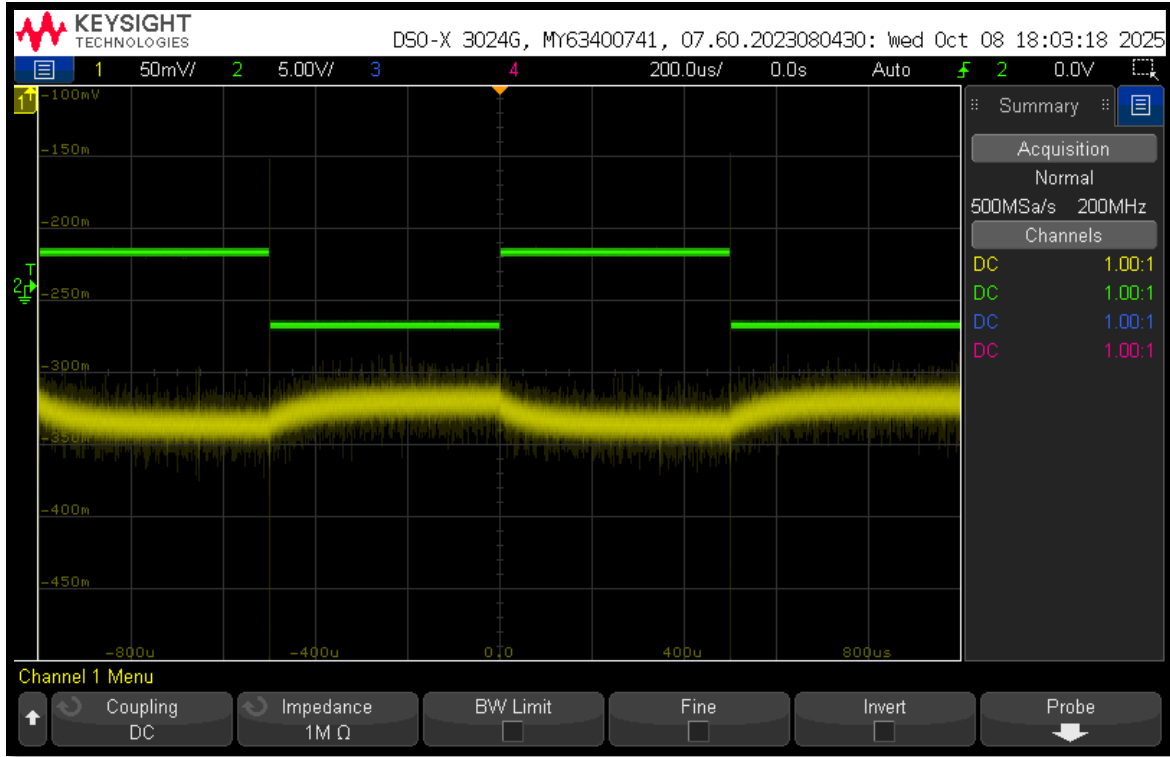


Figure 11: Green: Reduced 5.1V pk-to-pk amplitude, 50% duty cycle, Yellow: 1:1 SNR

The photodiode's output is a distorted square wave similar to the wave powering the LED. The peak-to-peak voltage was smaller by ~150 mV. It is clear the photodiodes rise and fall times are far more distorted and slower, nearing 50 ms. The desired 1:1 SNR ratio was found at a peak-to-peak of 5.1V.

To find the light intensity, the responsivity R is calculated at its maximum wavelength of $\lambda = 950$ nm. The responsivity is found in the data sheet to be $\sim R = 0.98$. Next, the open circuit voltage, V_{OC} , is measured by taking difference between on and off times of the 1:1 SNR wave. V_{OC} is determined to be ~150 mV. The short circuit current, I_{SC} , is estimated next using $n = 1.78$ from experiment 1. Then power can be calculated followed by intensity.

$$I_{SC} = I_L = (I_0) \left(e^{\frac{qV_{OC}}{nk_bT}} - 1 \right) = (2 \text{ nA}) \left(\frac{0.150}{e^{(1.78)(0.02585)}} - 1 \right) \approx 50 \text{ nA}$$

$$P_{On\ Diode} = \frac{I_{SC}}{R} = \frac{50 \times 10^{-9} A}{0.98} \approx 5.1 \times 10^{-8} W$$

$$E = \frac{P}{A} = \frac{5.1 \times 10^{-8} W}{(0.075\ cm^2)} \approx 6.8 \times 10^{-4}\ mW/cm^2$$

Concept Questions:

The estimated light intensity is significantly lower than any light source. Especially compared to sunlight which is 100 mW/cm². That is ~147,058.8235 times larger than the photodiode's measured light intensity.

Experiment 3:

The goal of experiment 3 is to make an optocoupler. A device that wirelessly transmits a signal from an LED to a photodetector. A pink IR LED is used to illuminate a OP999 PIN silicon photodiode. Similar to the previous experiments, the IR LED is angled directly at the OP999 photodiode. To ensure the most direct signal, the IR LED and photodiode were taped together. The LED was powered using a waveform generator outputting a 7V pk-to-pk amplitude, 1 kHz frequency, and a 50% duty cycle square wave. An oscilloscope is used to measure the wave.

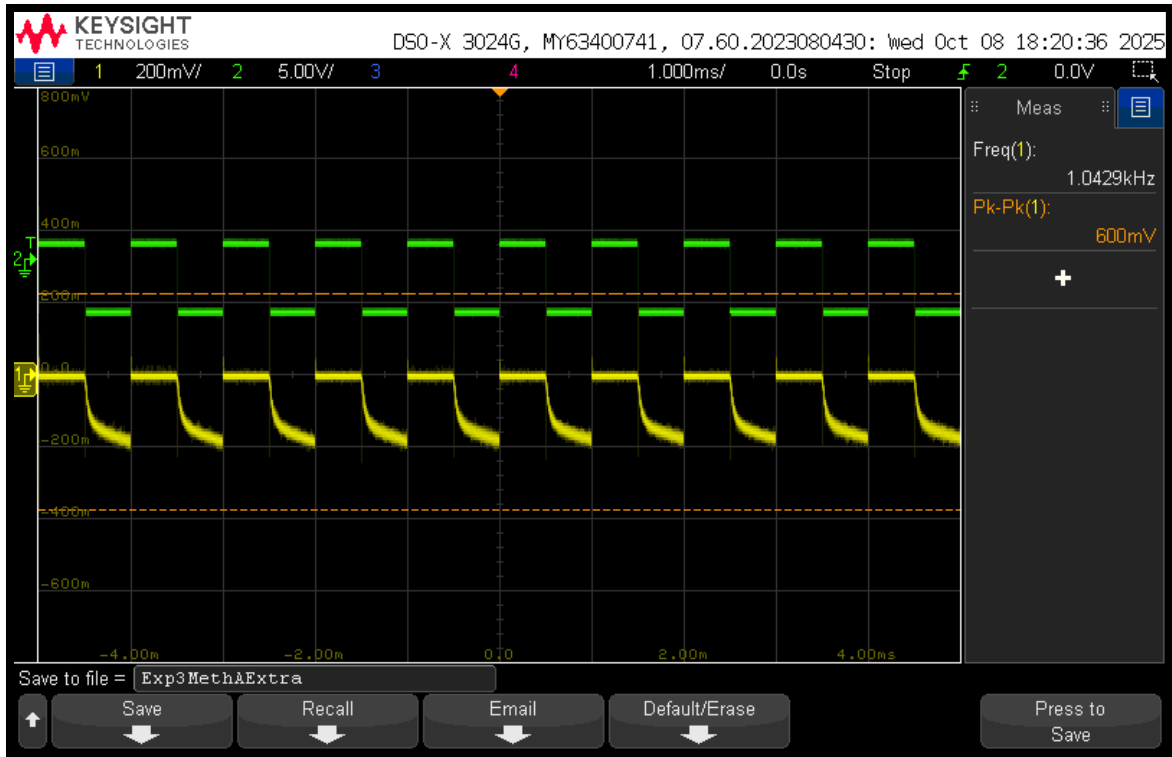


Figure 12: Green: Input Square Wave 1 kHz, 7V pk-to-pk, 50% duty cycle Yellow: Photodiode

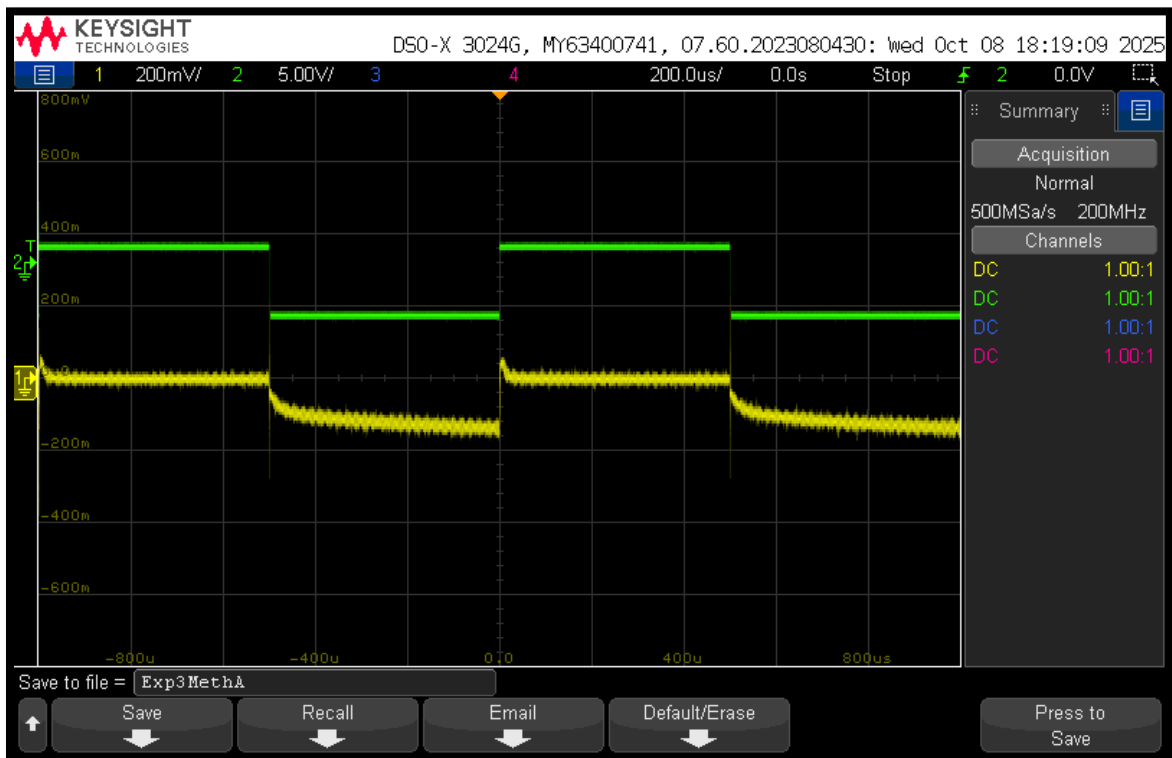


Figure 13: Green: Reduced 5V pk-to-pk input waveform r, Yellow: output voltage on photodiode

Using a higher voltage wave, a better resolution of the input waveform is captured. For the 7V peak-to-peak wave, the photodiode output looks like the input waveform but with a slower rise and fall time (~ 60 ms) and with more noise. For *Figure 13*, the reduced peak-to-peak waveform, it was harder to achieve a more consistent result. The output wave across the photodiode is still similar to the input square wave. But like before, it has a slower rise and fall time and more noise. The frequency and duty cycle are still the same, however. It is clear though that the lower peak-to-peak voltage leads to a worse signal to noise ratio.

Concept Questions:

It was difficult to acquire a suitable SNR. The distance between the transmitting LED and receiving photodiode led to an extremely sensitive output. Even when taped together, the two had to be perfectly fixed in place. A signal was only able to be measured when the LED and photodiode were in direct contact. At a lower voltage it was very difficult. At the higher voltage, it was still tedious but there was always some output. Without the tape to hold the two in place and while also holding them at the perfect position, it would be impossible to determine if what was being measured was an actual signal or just noise.

Conclusion:

This lab taught me how a photodiode's behavior can vary based on the provided illumination and bias configuration. By measuring the I-V characteristics in multiple scenarios, I understand that a photodiode can behave like a normal diode in the dark but will produce a current when in reverse bias under a light source. Fitting the data to the non-ideal diode equation explained the saturation current and ideality factor. As well as the linear and exponential behaviors. I learned to utilize the SMU and the waveform generator to take precise

measurements. A large takeaway is to always double check the measurement data before leaving lab. Or issues can occur like the one present in experiment one. Overall, I am more comfortable using these tools to learn about semiconductor behavior and how sensitive photodiodes are to the surrounding environment.

GenAI Usage:

AI was used to help with my plots and scale them correctly. With so many data points I could no longer manually enter all the data in Matlab. ChatGPT was the AI engine used for this as it explained how to import a .csv file. The prompts utilized were simple. I would provide my code from previous labs and ask it to make an edit to allow an import of a .csv file. It explained what expected behavior my data should show given the experiment setup. This is how I realized I was missing data from experiment one.

Citations:

“Photodiode – Working, Types, Characteristics & Applications.” *Electrical Technology*, 20 July 2022, www.electricaltechnology.org/2022/07/photodiode.html